PUBG Winner Prediction

Machine learning analysis of player behavior, combat performance, and survival strategies to predict match outcomes in PlayerUnknown's Battlegrounds.

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Dataset Overview

4.4M

Records

Total gameplay data points

29

Features

Gameplay metrics analyzed

200K

Samples

Used for model development

1.0

Target Range

winPlacePerc (0.0–1.0)

Data split: 70% Training | 15% Validation | 15% Testing

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Data Preprocessing & Engineering

Preprocessing Steps

- Removed inconsistent records (kills > 0 with zero damage)
- Treated extreme outliers using 99.5th percentile clipping
- Encoded categorical variable matchType
- Applied feature scaling for linear models

Feature Engineering

- totalDistance (walk + ride + swim)
- killEfficiency (kills / damageDealt)
- headshotRate (headshotKills / kills)
- totalHealing (combined healing features)
- isAggressive (behavioral indicator)

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Feature Correlation Analysis

The heatmap reveals strong positive correlations between kills and numerous gameplay features including killStreaks, killPlace, killPoints, DBNOs, walkDistance, weaponsAcquired, headshotKills, longestKill, matchDuration, maxPlace, rankPoints, rideDistance, roadKills, swimDistance, teammateKills, walkPoints, and winPlacePerc.

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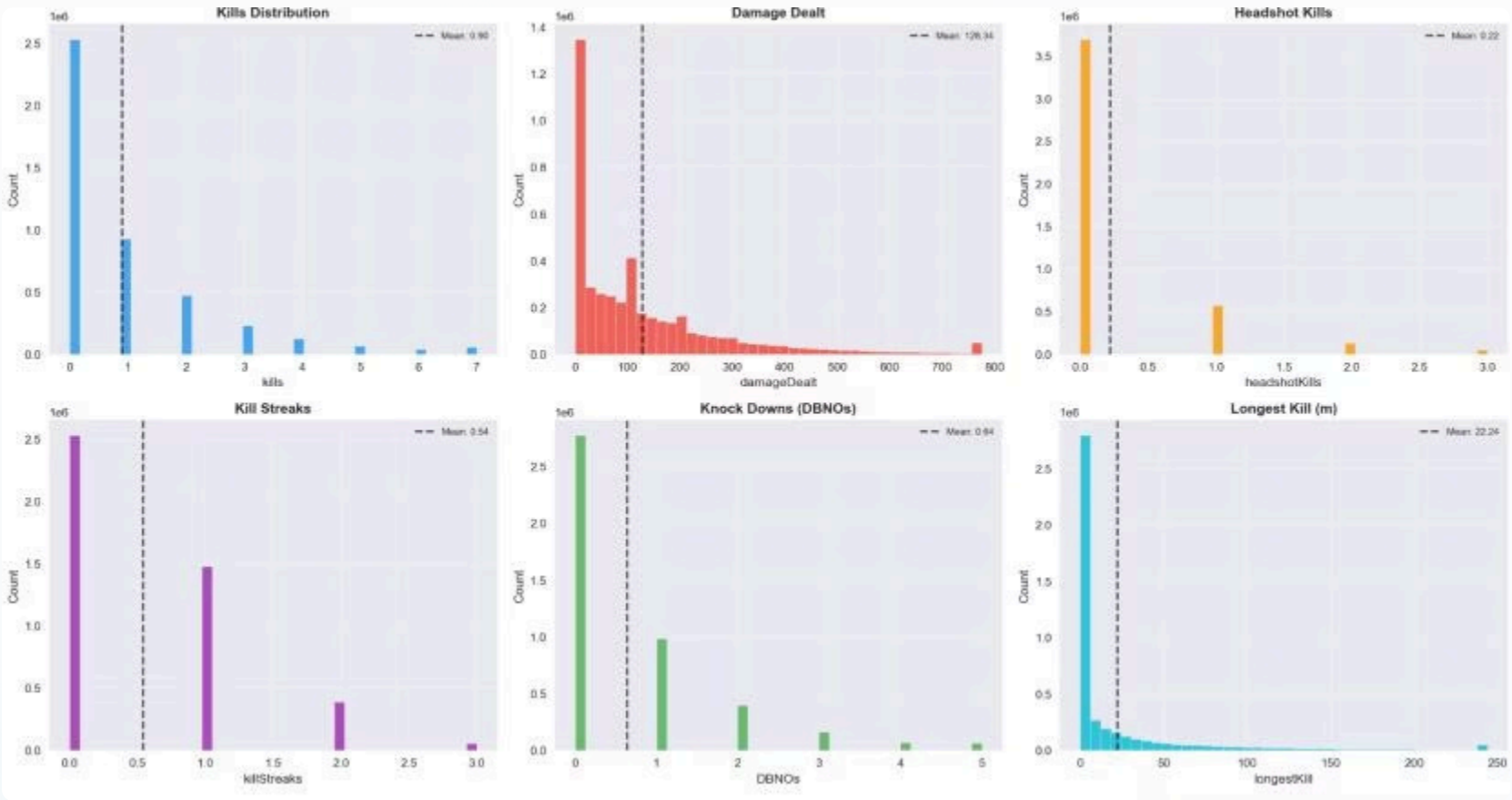
Combat Features Distribution



0.98

Kills

Mean value

128

Damage Dealt

Mean value

0.22

Headshot Kills

Mean value

0.84

Kill Streaks

Mean value

0.94

Knock Downs

Mean value

22

Longest Kill

Mean in meters

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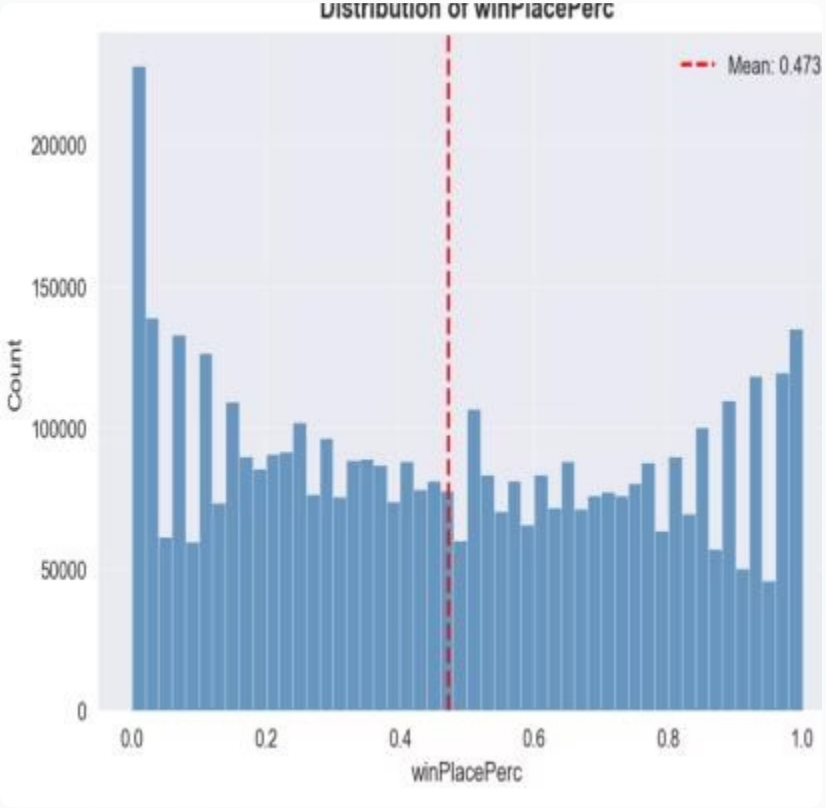
Target Variable: winPlacePerc



Distribution Characteristics

Right-skewed distribution with peak at 0.0. Mean value: 0.473

Box Plot Summary

- Minimum: 0.0
- Q1: 0.20
- Median: 0.45
- Q3: 0.65
- Maximum: 0.95

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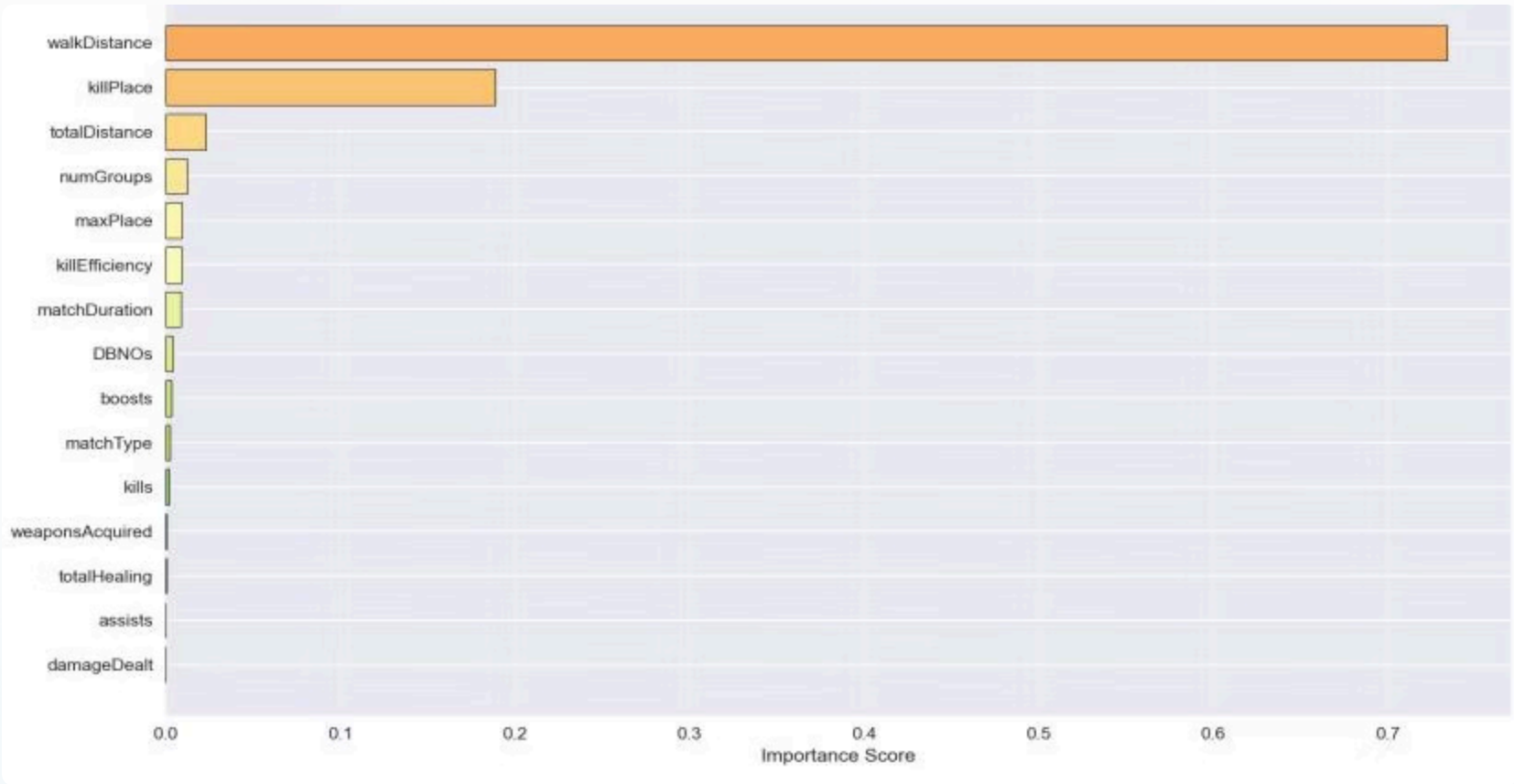
Top Feature Importances

Random Forest analysis revealed the most influential predictors of win placement.



1

walkDistance

0.72 importance score

2

killPlace

0.18 importance score

3

totalDistance

0.02 importance score

4

numGroups

0.015 importance score

5

maxPlace

0.01 importance score

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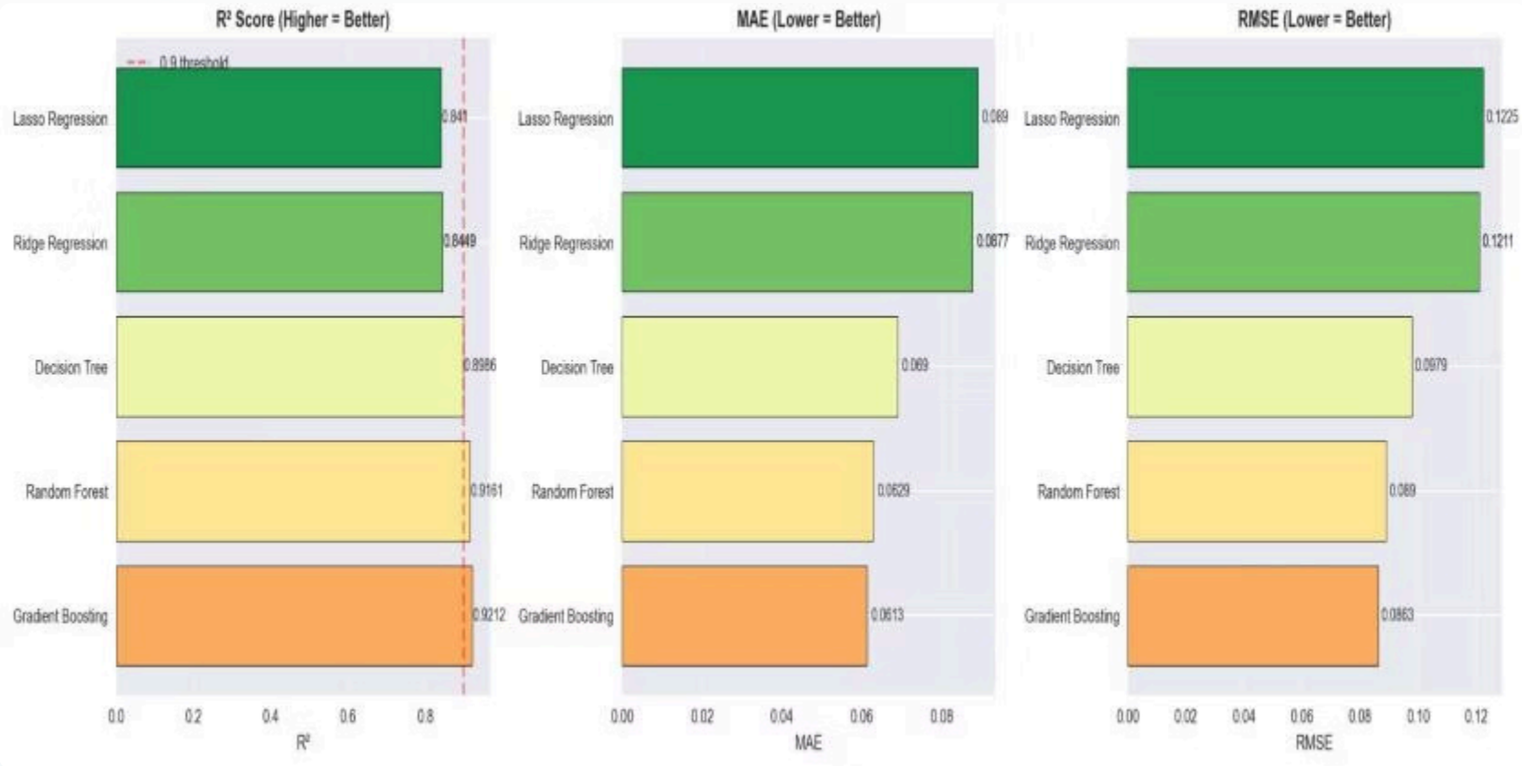
Model Comparison: R^2 Scores



Gradient Boosting

0.932 R^2 score

Lasso Regression

0.941 R^2 score

Random Forest

0.918 R^2 score

Decision Tree

0.866 R^2 score

Ridge Regression

0.849 R^2 score

All models exceeded the 0.8 threshold, with Gradient Boosting achieving the best balance of performance metrics.

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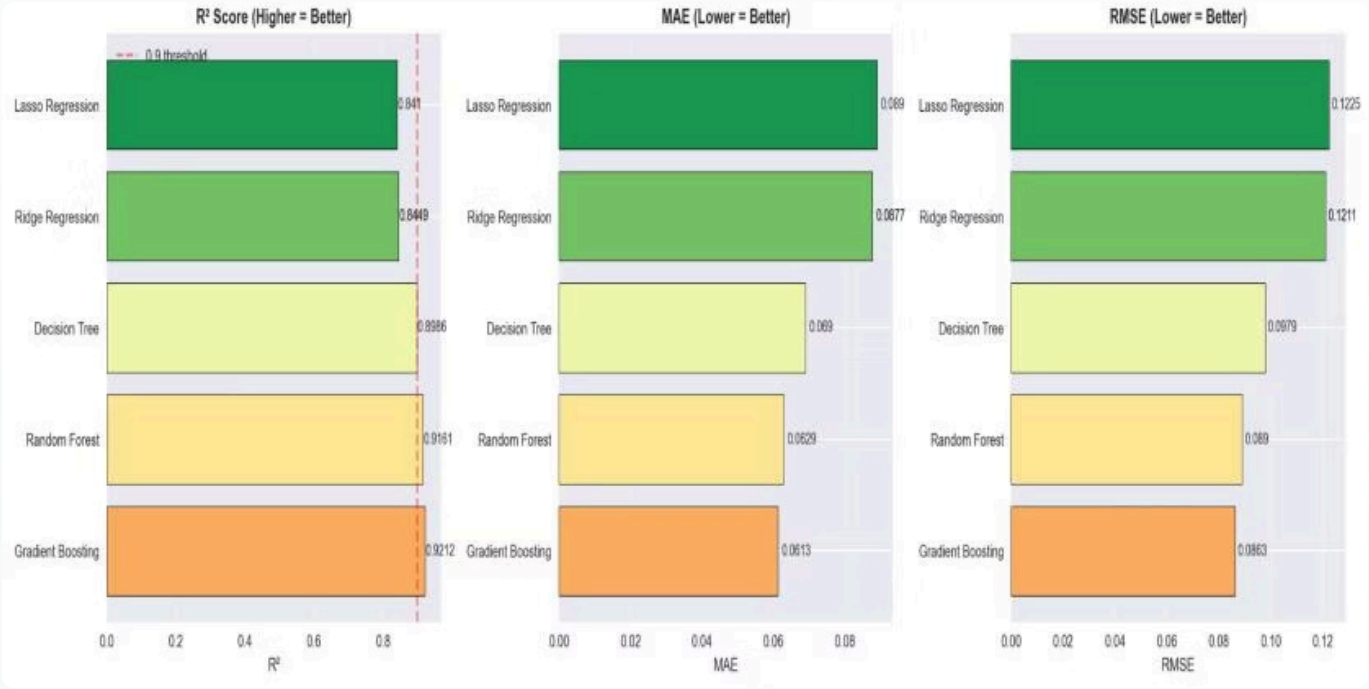
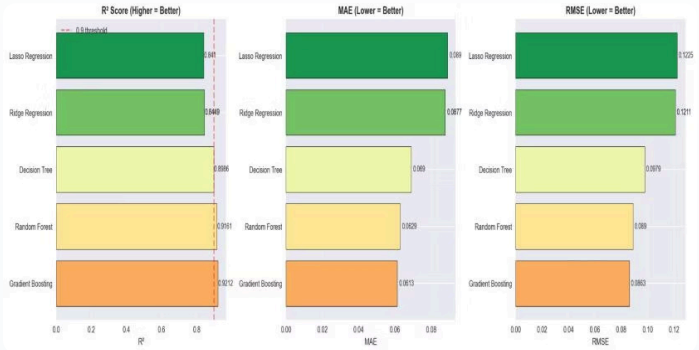
Model Comparison: Error Metrics

Mean Absolute Error (MAE)



Gradient Boosting achieved the lowest MAE at 0.0613, outperforming Lasso Regression (0.069), Ridge Regression (0.0877), Decision Tree (0.099), and Random Forest (0.1629).

Root Mean Squared Error (RMSE)



Decision Tree achieved the lowest RMSE at 0.0879, followed by Gradient Boosting (0.0883), Random Forest (0.099), Lasso Regression (0.1225), and Ridge Regression (0.1211).

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Key Findings & Conclusions

Best Model

Gradient Boosting achieved the best balance with R^2 of 0.932, MAE of 0.0613, and RMSE of 0.0883

Key Predictor

walkDistance emerged as the most important feature with 0.72 importance score

Target Distribution

winPlacePerc showed right-skewed distribution with mean of 0.473 and median of 0.45

The analysis successfully identified key gameplay factors influencing match outcomes, with movement and positioning proving most critical to victory.

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